DIPLOMA PROGRAMME PRACTICE

Mathematics

Applications and interpretation SL

Set A / Paper 1 / Short response

90 minutes · 80 marks · All questions compulsory

Use a graphic display calculator and a clean Mathematics formula booklet.

Write your reasoning and working in the spaces provided; use additional answer paper if needed. Unless stated otherwise, give exact answers or round numerical results to three significant figures.

Diagrams are illustrative unless coordinates or dimensions are specified.

QUESTION 1

6 marks

A rectangular glass panel is recorded as 1.24 m by 0.86 m. Each length was rounded to the nearest 0.01 m.

(a) State the lower and upper bounds for the longer side, with appropriate inequalities. [2]

(b) A frame needs at least 1.060 m² of glass. Determine whether the recorded dimensions guarantee that it will fit this area requirement. [4]

QUESTION 2

6 marks

In the first week of a trial a library issues 160 passes. In each subsequent week it issues 12 more passes than in the previous week.

(a) Find the number issued in week 9. [2]

(b) Find the first week by the end of which the total number issued exceeds 2000. [4]

QUESTION 3

6 marks

A cold-store probe gives a reading of 2.6 V at −4 °C and 5.0 V at 8 °C. Assume that voltage is a linear function of temperature over this interval.

(a) Find the voltage model V = mT + c. [3]

(b) A reading of 4.3 V is observed. Estimate the temperature and explain why this is interpolation. [3]

QUESTION 4

6 marks

A rotating sprinkler waters a sector with radius 7.5 m and central angle 128°. Within the sector, water lands uniformly at 3.0 litres per square metre during one cycle.

(a) Calculate the length of the curved outer edge. [2]

(b) A 190-litre tank is full. Decide whether it can supply one cycle, showing the unrounded calculation used to decide. [4]

QUESTION 5

6 marks

Seven randomly selected morning journeys took 12, 13, 13, 14, 15, 17 and 28 minutes. Use the median-of-halves convention, excluding the overall median.

(a) Find the interquartile range and determine whether 28 is an outlier using the 1.5 × IQR rule. [4]

(b) A manager uses the sample mean to advertise a typical journey at any time of day. Give two distinct reasons for caution. [2]

QUESTION 6

6 marks

A parcel desk processes 300 items: 180 are standard and 120 are express. Of the standard items, 9 arrive late; of the express items, 12 arrive late. One processed item is selected at random.

(a) Find the probability that the item arrived late. [2]

(b) Given that it arrived late, find the probability that it was express. [2]

(c) Determine whether being express and arriving late are independent for these data. [2]

QUESTION 7

7 marks

For six bus stops the shelter length x (m) and the number y of seats are paired as follows: (2, 4), (3, 5), (4, 7), (5, 8), (6, 11), (7, 13).

(a) Use technology to find the least-squares regression line of y on x and the value of Pearson's r. [4]

(b) For the stop with x = 5, find the residual (observed minus predicted). Explain its sign. [3]

QUESTION 8

7 marks

At a loading bay, each crate independently has probability 0.12 of needing a second scan. A batch contains 18 crates.

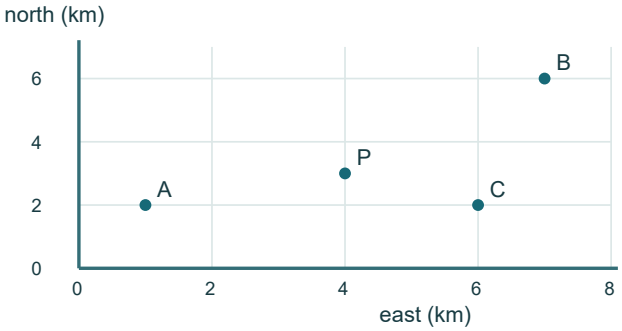
(a) Calculate the probability that exactly two crates need a second scan. [3]

(b) Extra staff are called when three or more crates need a second scan. Find the probability of calling them, and explain why a change of scanner during a batch could invalidate the model. [4]

QUESTION 9

7 marks

Two repair depots are located at A(1, 2) and B(7, 6), in kilometres. Travel is modelled by straight-line distance. The boundary contains points equidistant from A and B.



(a) Determine the equation of the boundary. [3]

(b) A new site C(6, 2) opens. A fault is reported at P(4, 3). Determine whether adding C changes which depot serves P. [4]

QUESTION 10

7 marks

The amount of water in a tank is $V(t) = 24 + 12t - 1.5t^2$ litres, for $0 \le t \le 6$ minutes.

(a) Find the rate of change at $t = 2$, including units. [3]

(b) Determine the maximum amount of water during the modelled period and justify that it is a maximum. [4]

QUESTION 11

8 marks

A co-operative borrows 9600 dollars. Interest is 6% nominal per annum compounded monthly. It makes 36 equal payments at the end of each month; there are no fees and the final balance is zero.

(a) Determine the monthly payment to the nearest cent. [3]

(b) Find the total interest paid to the nearest dollar, using the unrounded payment. [2]

(c) The co-operative considers 48 payments at the same rate instead. Compare the monthly payment and the total interest, explaining the difference. [3]

QUESTION 12

8 marks

Outflow from a sluice is measured every two minutes. At t = 0, 2, 4, 6 and 8 minutes the rates are 18, 25, 29, 26 and 20 litres per minute respectively.

(a) Use the trapezoidal rule with all measurements to estimate the discharged volume. [3]

(b) A technician instead uses only the first, middle and last readings. Calculate the revised estimate and its percentage difference relative to the five-reading estimate. [4]

(c) Explain why neither estimate establishes the exact volume. [1]

MARKING AND REVIEW

Worked solutions

Applications and interpretation SL

Set A / Paper 1 / Short response

Maximum mark: 80. The allocations are for independent self-assessment.

M = method, A = accuracy or stated result, R = reasoning. Award equivalent valid methods. Follow through from an earlier numerical slip when the subsequent method is correct and the task is not simplified.

Keep unrounded calculator values until the final answer. Accept sensible equivalent accuracy unless the question specifies otherwise.

Read the student's written interpretation as well as their numerical result. A calculator output alone may not justify a conclusion.

SOLUTION 1

6 marks

(a) State the lower and upper bounds for the longer side, with appropriate inequalities. [2]

A1 Lower bound 1.235 m.

A1 Upper bound 1.245 m is excluded: $1.235 \le L < 1.245$.

(b) A frame needs at least 1.060 m² of glass. Determine whether the recorded dimensions guarantee that it will fit this area requirement. [4]

M1 Use both lower bounds for the minimum possible area.

A1 $1.235 \times 0.855 = 1.055925$ m².

R1 This is less than 1.060 m², even though the product of the recorded lengths exceeds 1.060.

R1 No guarantee: some panels consistent with the measurements are too small.

SOLUTION 2

6 marks

(a) Find the number issued in week 9. [2]

M1 Use $160 + (9 - 1) \times 12$.

A1 256 passes.

(b) Find the first week by the end of which the total number issued exceeds 2000. [4]

M1 Sum the arithmetic sequence: $S(n) = n[320 + 12(n - 1)]/2$.

A1 $S(9) = 1872$.

A1 $S(10) = 2140$.

R1 Week 10 is the first because the cumulative total strictly increases.

SOLUTION 3

6 marks

(a) Find the voltage model V = mT + c. [3]

M1 m = (5.0 − 2.6)/(8 − (−4)).

A1 m = 0.20 V per °C.

A1 Substitution gives c = 3.4 V.

(b) A reading of 4.3 V is observed. Estimate the temperature and explain why this is interpolation. [3]

M1 Solve 4.3 = 0.20T + 3.4.

A1 T = 4.5 °C.

R1 The voltage lies between 2.6 and 5.0 V, the measured endpoints.

SOLUTION 4

6 marks

(a) Calculate the length of the curved outer edge. [2]

M1 Take 128/360 of the circumference 2π × 7.5.

A1 Arc length = 16.8 m (3 sf).

(b) A 190-litre tank is full. Decide whether it can supply one cycle, showing the unrounded calculation used to decide. [4]

M1 Sector area = (128/360)π × 7.5².

A1 Area = 20π m².

A1 Required volume = 60π = 188.496 litres.

R1 Yes; approximately 1.50 litres remain, before any unmodelled losses.

SOLUTION 5

6 marks

(a) Find the interquartile range and determine whether 28 is an outlier using the 1.5 × IQR rule. [4]

A1 Q1 = 13 and Q3 = 17.

A1 IQR = 4 minutes.

M1 Upper fence = 17 + 1.5 × 4 = 23.

A1 28 exceeds 23, so it is an outlier.

(b) A manager uses the sample mean to advertise a typical journey at any time of day. Give two distinct reasons for caution. [2]

R1 The long journey raises the mean relative to the median; the mean may be unrepresentative.

R1 Only morning journeys were sampled, so other times of day are not represented.

SOLUTION 6

6 marks

(a) Find the probability that the item arrived late. [2]

M1 Combine the two late groups: 9 + 12.

A1 21/300 = 0.0700.

(b) Given that it arrived late, find the probability that it was express. [2]

M1 Restrict the denominator to the 21 late items.

A1 12/21 = 4/7.

(c) Determine whether being express and arriving late are independent for these data. [2]

A1 P(late given express) = 12/120 = 0.10.

R1 This differs from 0.07, so the events are not independent.

SOLUTION 7

7 marks

(a) Use technology to find the least-squares regression line of y on x and the value of Pearson's r. [4]

M1 Enter paired data and select a linear regression predicting seats from length.

A1 Slope = 1.83.

A1 Intercept = -0.229; use y = mx + b.

A1 r = 0.988.

(b) For the stop with x = 5, find the residual (observed minus predicted). Explain its sign. [3]

M1 Predict at x = 5 using the unrounded fitted coefficients.

A1 Residual = -0.914 seats.

R1 The actual stop has fewer seats than its fitted value.

SOLUTION 8

7 marks

(a) Calculate the probability that exactly two crates need a second scan. [3]

M1 Use X distributed as binomial with n = 18 and p = 0.12.

M1 Evaluate P(X = 2) using the binomial probability function.

A1 0.285.

(b) Extra staff are called when three or more crates need a second scan. Find the probability of calling them, and explain why a change of scanner during a batch could invalidate the model. [4]

M1 Use the complement of X ≤ 2.

A1 P(X ≤ 2) = 0.631.

A1 P(call) = 0.369.

R1 The two scanners may have different second-scan probabilities, violating the constant-p assumption.

SOLUTION 9

7 marks

(a) Determine the equation of the boundary. [3]

A1 Midpoint = (4, 4).

M1 The slope of AB is 4/6, so the perpendicular slope is −3/2.

A1 $y - 4 = -1.5(x - 4)$, or $y = -1.5x + 10$.

(b) A new site C(6, 2) opens. A fault is reported at P(4, 3). Determine whether adding C changes which depot serves P. [4]

M1 Compare squared distances from P to A and B.

A1 $PA^2 = 10$ and $PB^2 = 18$, so A originally serves P.

A1 $PC^2 = 5$.

R1 C is closer than both original depots, so the assignment changes to C.

SOLUTION 10

7 marks

(a) Find the rate of change at t = 2, including units. [3]

M1 Differentiate term by term.

A1 $V'(t) = 12 - 3t$.

A1 $V'(2) = 6$ litres per minute.

(b) Determine the maximum amount of water during the modelled period and justify that it is a maximum. [4]

M1 Set the derivative to zero.

A1 t = 4 minutes, inside the stated interval.

A1 $V(4) = 48$ litres.

R1 The derivative is positive before 4 and negative afterwards (or the parabola opens down).

SOLUTION 11

8 marks

(a) Determine the monthly payment to the nearest cent. [3]

M1 Monthly rate = 0.06/12 = 0.005.

M1 Use 36 end-of-month payments and present value 9600 with zero final value.

A1 Payment = 292.05 dollars per month.

(b) Find the total interest paid to the nearest dollar, using the unrounded payment. [2]

M1 Total paid = 36 × the unrounded payment.

A1 Subtract 9600 to give 914 dollars interest.

(c) The co-operative considers 48 payments at the same rate instead. Compare the monthly payment and the total interest, explaining the difference. [3]

A1 The monthly payment is lower.

A1 Total interest is higher.

R1 Principal is outstanding for longer, so interest accrues over more periods despite the smaller instalments.

SOLUTION 12

8 marks

(a) Use the trapezoidal rule with all measurements to estimate the discharged volume. [3]

M1 Width = 2 min; count each internal ordinate twice.

M1 Use 2/2 × [18 + 2(25 + 29 + 26) + 20].

A1 198 litres.

(b) A technician instead uses only the first, middle and last readings. Calculate the revised estimate and its percentage difference relative to the five-reading estimate. [4]

M1 With width 4 the estimate is 4/2 × (18 + 2 × 29 + 20).

A1 192 litres.

M1 Use 100 × (198 − 192)/198.

A1 The revised estimate is 3.03% lower.

(c) Explain why neither estimate establishes the exact volume. [1]

R1 Interpolation between measured rates assumes straight segments; the true rate can follow a different curve.